3/1

12.1 Counting

Multiplication Principle

Permutations] order matters $\quad {}_nP_k = \frac{n!}{(n-k)!}$

Combinations] order doesn't matter $\quad {}_nC_k = \frac{n!}{k!(n-k)!}$

5 books

A B C D E

Arrange 3 on a shelf

A B C | E D

"

A B C | D E

. . .

$\underset{1}{A}\ \underset{2}{B}\ \underset{3}{C} \neq \underset{1}{C}\ \underset{2}{A}\ \underset{3}{B}$

$\frac{5!}{2!}$ ← ways to arrange 5 distinct books

↖ ways to arrange the last 2 books

. . . $\underset{1}{D}\ \underset{2}{C}\ \underset{3}{A} = {}_5P_3 = \frac{5!}{2!}$

5 books

A B C D E

Arrange 3 on a shelf

A B C E D

"

A B C D E

$\underset{1}{A}\ \underset{2}{B}\ \underset{3}{C} \neq \underset{1}{C}\ \underset{2}{A}\ \underset{3}{B}$ ...

$\frac{5!}{2!}$ ← ways to arrange 5 distinct books

↖ ways to arrange the last 2 books

... $\underset{1}{D}\ \underset{2}{C}\ \underset{3}{A} = {}_5P_3 = \frac{5!}{2!}$

5 books

A B C D E

Choose any 3 books

A B C = C A B

$\binom{5}{3} = \frac{5!}{3!\,2!}$ ← ways to arrange books not chosen

↑ ways to arrange 3 chosen books

${}_nP_k \geq {}_nC_k$

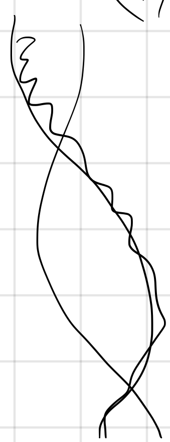
12.1

1. 5 fertilizer
   2 light levels
   4 replicates

$$\underset{\text{fertilizer}}{\underline{5}} \cdot \underset{\text{light levels}}{\underline{2}} \cdot \underset{\text{replicates}}{\underline{4}} = 40$$

2. 9749 nucleotides

A(T/U)CG   9749·4 genomes



AT
≠
TA

12.1.4

Example 8

MISSISSIPPI

$$\_\ \overset{\substack{2\\1}}{I}\ \_\ \overset{\substack{1\\2}}{I}\ \_\ \_\ \_\ \overset{\substack{3\\3}}{I}\ \overset{\substack{4\\4}}{I}\ \_\ \_$$

M – 1

I – 4

S – 4

P – 2

$$\frac{11!}{\underbrace{1!}_{\text{ways to arrange 1 M}}\ \underbrace{4!}_{\text{ways to arrange 4 S's}}\ 4!\ 2!}$$

$11!$ ← ways to arrange 11 letters

$$\underbrace{\binom{11}{1}}_{M}\underbrace{\binom{10}{4}}_{I}\underbrace{\binom{6}{4}}_{S}\underbrace{\binom{2}{2}}_{P} = \frac{11!}{1!\,\cancel{10!}}\cdot\frac{\cancel{10!}}{4!\,\cancel{6!}}\cdot\frac{\cancel{6!}}{4!\,\cancel{2!}}\cdot\frac{\cancel{2!}}{2!\,0!} = \frac{11!}{1!4!4!2!}$$

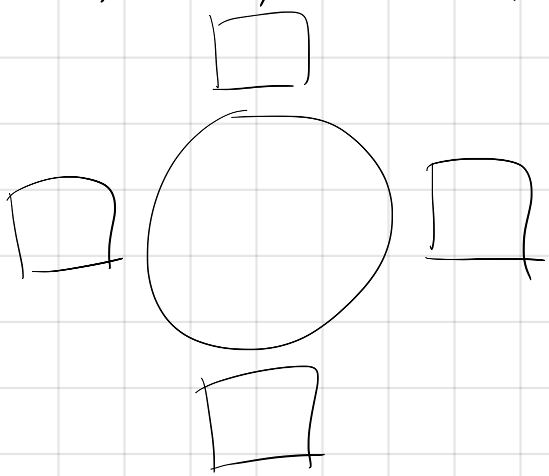
ways to choose 1 M in 11 spots

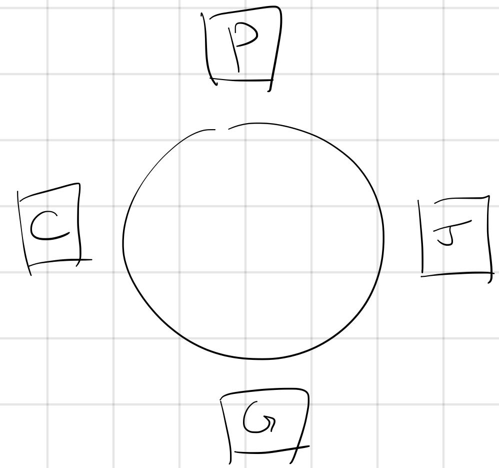
ways to choose 4 I's in remaining 6 spots

Note:

$0! = 1$

36. Paula, Cindy, Gloria, Jenny

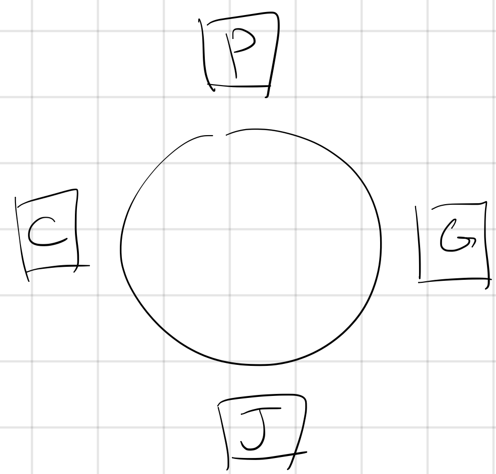




Cindy must sit to the left of Paula.

2 if seat number does not matter

4·2 if seat number matters



Binomial Theorem

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$$

$$(x+y)^0 = 1$$

$$(x+y)^1 = x+y$$

$$(x+y)^2 = x^2+2xy+y^2$$

$$(x+y)^3 = x^3+3x^2y+3xy^2+y^3$$

$$
\begin{array}{ccccccccc}
 & & & & 1 & & & & \\
 & & & 1 & & 1 & & & \\
 & & 1 & & 2 & & 1 & & \\
 & 1 & & 3 & & 3 & & 1 & \\
 & \binom{3}{0} & & \binom{3}{1} & & \binom{3}{2} & & \binom{3}{3} & \\
1 & & 4 & & 6 & & 4 & & 1 \\
\binom{4}{0} & & \binom{4}{1} & & \binom{4}{2} & & \binom{4}{3} & & \binom{4}{4}
\end{array}
$$

Expand $(2x-3y)^5$

$$\binom{5}{0}\cdot(2x)^5(-3y)^0+\binom{5}{1}\cdot(2x)^4(-3y)+\binom{5}{2}(2x)^3(-3y)^2$$
$$+\binom{5}{3}(2x)^2(-3y)^3+\binom{5}{4}(2x)(-3y)^4+$$
$$\binom{5}{5}(2x)^0(-3y)^5$$

Count the number of subsets of $\{a,b,c\}$

$\{\,\}$ $\quad \binom{3}{0}$

$+$

$\{a\},\{b\},\{c\}$ $\quad \binom{3}{1}$

$+$

$\{a,b\},\{b,c\},\{a,c\}$ $\quad \binom{3}{2}$

$+$

$\{a,b,c\}$ $\quad \binom{3}{3}$

$$\sum_{k=0}^{3}\binom{3}{k}\cdot 1^{3-k}\cdot 1^{k}=(1+1)^3=2^3$$

$$\sum_{k=0}^{n}\binom{n}{k}=(1+1)^n=2^n$$

47. poker

52 cards

4 suits

13 numbers

$$\frac{\binom{52}{4}}{\underbrace{\text{ways to choose not 2 pairs}}}$$

$52 \cdot 48 \cdot 3 \cdot 3$